Reverse Universe Theory as a Regularized Late-Time Cosmology Sector

Sections 1–4 Draft • Empirical Status, Enlarged-r_d Derivation Path, and Operational Framing

1. Abstract

We report the current empirical status of a regularized late-time sector within the Reverse Universe Theory (RUT) framework. In the present BAO+growth workflow, the healthiest anchored solutions are best interpreted not as a universal all-redshift deformation, but as a statistically competitive late-time correction sector occupying a broad parameter basin near $\mathbf{d_G \approx 0.18–0.19}$, $\boldsymbol{\delta a \approx 0.04–0.05}$, and $\mathbf{z_{cut} \approx 0.8–1.0}$. Its strongest repeated empirical signature is a structured growth-to-geometry transfer effect rather than a uniform improvement across observables.

A second focus of the note is the enlarged effective drag horizon. Earlier work identified a stabilized interior preference near $\mathbf{r_d \approx 181.75}$, suggesting that the preferred horizon shift is not well described as simple boundary chasing. The present framework now pushes that signal through three levels: candidate mechanism, structured-transport derivation path, and a first closed minimal model in which the same organized primordial sector affects both transport and background expansion.

The fair current claim is narrow. RUT is not presented here as a demonstrated replacement for standard cosmology across all redshifts, nor as a completed first-principles early-plasma derivation. Rather, this draft argues that a regularized late-time bridge with a repeatable empirical fingerprint and an operational enlarged-r_d program now deserves compact, reviewer-facing presentation and direct falsification.

2. Executive empirical summary

The strongest current empirical reading of RUT is more disciplined than earlier broad formulations. The regularized bridge is now best understood as a late-time correction sector rather than a universal deformation of the entire expansion history.

In the present workflow, its recurring healthiest region sits near $\mathbf{d_G \approx 0.18–0.19}$, $\boldsymbol{\delta a \approx 0.04–0.05}$, with activation centered near $\mathbf{a^* = 0.50}$ and strongest anchored performance in the approximate range $\mathbf{z_{cut} \approx 0.8–1.0}$. The best anchored model-selection result reported in the present stream is $\mathbf{\Delta BIC \approx +0.390}$ at $\mathbf{z_{cut} = 0.8}$.

Just as important as the nominal best point is the basin structure. The present signal is no longer strongest as a seed-fragile optimizer spike. It occupies a broad, smooth competitive plateau. That matters because it shifts the interpretation from “single lucky fit” toward “repeatable late-time sector behavior under tightening.”

The empirical fingerprint is structured rather than uniform. The bridge repeatedly improves selected radial BAO observables and several growth measurements while repeatedly paying cost in some lower-redshift and more transverse-like BAO observables. The cleanest current reading is therefore not global replacement, but a repeatable growth-to-geometry transfer signature with an identifiable domain of usefulness and an identifiable transfer weakness.

3. Bridge identity card

Field	Current best reading
Type	Regularized late-time correction sector
Status	Statistically competitive, but not decisive
Preferred region	d_G ≈ 0.18–0.19; δa ≈ 0.04–0.05; a* = 0.50; z_cut ≈ 0.8–1.0
Best anchored reading	Near-competitive with a broad basin rather than a single optimizer accident
Strongest asset	Repeatable growth-to-geometry transfer effect
Principal weakness	Low-z to high-z transfer, especially into broader BAO structure
Enlarged-r_d link	Progresses from stabilized signal to derivation path to first operational model
Current interpretation boundary	Not yet a demonstrated all-redshift replacement or full first-principles early-plasma derivation
Most important next test	Shared-region test: enlarge r_d while preserving broader benchmark behavior

The enlarged-horizon rule is sharper than before. In the current operational form, the larger horizon survives only when weighted drag-era transport enhancement exceeds weighted drag-era expansion backreaction. This is the strongest compact statement of what the bridge must actually do, rather than merely suggesting that extra structure “might help.”

4. Regularization arc and basin evidence

The empirical status of the bridge is strongest when read as a **regularization story**, not just a best-fit story. Earlier looser versions could improve raw fit by behaving too flexibly across redshift. The present regularized stream matters more because tightening did not erase the signal. Instead, it compressed the bridge into a narrower and healthier role: a late-time sector with a broad competitive basin and a repeatable transfer pattern.

The recurring preferred region now looks much more stable than a single optimizer accident. In the current BAO+growth workflow, the anchored late-time bridge remains healthiest near $\mathbf{d_G \approx 0.18\text{–}0.19}$, $\mathbf{\delta a \approx 0.04\text{–}0.05}$, and $\mathbf{z_{cut} \approx 0.8\text{–}1.0}$, with activation centered near **a* = 0.50**. The strongest anchored model-selection result reported in the present stream is $\mathbf{\Delta BIC \approx +0.390}$ at $\mathbf{z_{cut} = 0.8}$. This is not a decisive win over ΛCDM, but it is materially different from collapse under regularization.

Equally important is the basin shape. The present bridge no longer appears best described as a seed-fragile spike. The healthier reading is a **broad, smooth competitive plateau** across a constrained late-time region. That broadness matters because it makes the bridge easier to interpret physically: if a signal survives only at one numerically fragile point, it is hard to defend as structure; if it survives across a neighborhood of settings after tightening, it begins to look like a real empirical lane worth isolating and testing.

The observational pattern is also not random. The bridge’s gains are structured: it repeatedly improves selected radial BAO observables and several growth measurements, while repeatedly paying cost in some lower-redshift and more transverse-like BAO observables. That asymmetry is one of the strongest reasons to avoid describing the result as a numerical fluke. The more disciplined interpretation is that the bridge is picking up a specific **growth-to-geometry transfer fingerprint**, not offering a uniform all-purpose correction.

This is why the regularized bridge should presently be framed as a **late-time correction sector** rather than a full-history cosmology replacement. Its strongest present empirical role is local and structured: it helps in a repeatable late-time lane, occupies a broad healthy basin there, and weakens when asked to carry that same structure too far into broader all-redshift behavior. In reviewer-facing terms, that narrower claim is stronger because it matches what the present data actually support.

Compact summary of the regularization arc

- tightening reduced flexibility rather than improving rhetoric alone;
- the signal survived in a narrower domain;
- the preferred region is broad enough to resist a “single lucky fit” reading;
- the bridge now reads best as a repeatable late-time growth-to-geometry sector;
- the next scientific question is whether this same compact structure survives broader transfer tests and the enlarged-r_d consistency program.

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Sections 5-9 Draft • Enlarged Horizon, Closed Minimal r_d Model, Reviewer Pressure, Falsifiers, and Conclusion

5. Enlarged sound horizon: from stabilized signal to derivation path

One of the most important present RUT results is that the preferred enlarged sound horizon no longer looks best described as a simple boundary artifact. In the current benchmark stream, the rd190 soft-cap rerun preserves an interior optimum near $\mathbf{r_d \approx 181.75}$, with the local scan rising on both sides of the minimum. That changes the status of the enlarged horizon from a suspicious wall-chasing direction to a mechanism-facing signal that requires explanation.

The first disciplined interpretation is narrow. The claim is not that the early plasma has already been derived from first principles in completed form. The claim is that the preferred frozen horizon may not be purely acoustic in origin. Instead, it may reflect transport in a coupled gravito-electromagnetic-acoustic plasma, where organized radiative and field structure modifies the effective propagation scale imprinted into later observables.

The derivation path in the current draft stream now has three levels. First, the horizon is treated as a **candidate signal**: the enlarged preferred scale survives relaxation of the earlier cap and remains stabilized near $\mathbf{r_d \approx 181.75}$. Second, the horizon is given a **structured-transport derivation path**, where the standard drag-epoch horizon is generalized by replacing the purely acoustic speed with an effective structured transport speed. Third, the horizon enters a **first operational model** in which the same organized sector affects both transport and expansion.

This is the key scientific change in status. Earlier wording could stop at “the primordial plasma may have carried more than ordinary sound-like motion alone.” The stronger formulation now says exactly what must be true for the larger horizon to survive: the organized primordial sector must improve effective transport more than it increases effective expansion across the drag-era window. In compact form, that condition becomes $\mathbf{\langle\Delta_c\rangle_d > \langle\Delta_H\rangle_d}$.

This matters because it reduces the degree of freedom in how the enlarged horizon is discussed. The framework is no longer strongest when saying only that a larger r_d “fits better” or “looks preferred.” It is stronger when it says that the larger horizon follows from a partially closed transport-and-backreaction structure under explicit assumptions.

The present interpretation boundary should stay explicit. The enlarged r_d remains a large departure from the standard expectation, and that alone means it will attract scrutiny. The right posture is therefore not triumphal. It is to treat the enlarged horizon as an operational prediction path whose mechanism has been materially strengthened but not fully closed.

6. Closed minimal r_d model with backreaction

To move beyond a purely schematic enlarged-horizon story, the current physics stream introduces a closed minimal r_d model. Its purpose is narrow and practical: fix the key bridge ingredients, allow the same organized primordial sector to affect both transport and expansion, and make the enlarged-horizon condition quantitatively testable.

The first step is to lock the bridge choices. The model fixes a bounded phase-weight function $\Pi(\omega)$ and a bounded structured-radiation-flow factor Λ■$_L(z)$. These choices are not claimed as unique or final. Their role is to make the first-pass model positive, bounded, and disciplined enough to defend.

The organized transport sector is then written in compact form through an effective B_{eff}^2 term, so that the effective propagation law becomes a modified $c_{eff}^2(z)$. This is the first point at which the enlarged-horizon mechanism becomes more than verbal. The transport term is now explicitly linked to phase weighting, radiation-flow weighting, and organized dual-sector structure.

The same organized sector is then allowed to contribute to the expansion through an effective energy density $\rho_T(z)$ and pressure $p_T(z)$, so that $H_{RUT}^2(z)$ includes a matching backreaction term. This is the crucial hardening step. The enlarged horizon is no longer allowed to borrow structure only in the transport term while leaving the expansion side external. The same sector must now pay its own backreaction cost.

With these ingredients, the closed horizon equation becomes a first minimal form in which one organized primordial sector both raises the effective propagation scale and backreacts on the expansion rate. The first-order condition derived from this model is especially important: the enlarged horizon survives when weighted drag-era transport enhancement exceeds weighted drag-era expansion backreaction.

The model also keeps the correspondence limit explicit. If the added sector decouples, then c_{eff}^2 returns to c_s^2, the backreaction term vanishes, H_{RUT} returns to H_{base}, and r_d^{RUT} returns to the baseline horizon. The fair reading remains narrow: this is not yet a fully evolved field system, but it is a real upgrade from a mechanism sketch to a first operational model.

7. Reviewer-facing pressure points: BIC, H0, and large r_d

The present RUT materials are strongest when they confront the main reviewer-facing objections directly rather than letting them appear only indirectly through cautious wording. Three pressure points matter most: model-selection discipline, the H_0 issue, and the scale of the enlarged r_d itself.

7.1 BIC and parameter-count discipline. The current empirical position is near-competitive, not decisive. The healthiest anchored late-time bridge region is reported with a best anchored model-selection result of **ΔBIC ≈ +0.390** at **z_{cut} = 0.8**. That is important because it shows that tightening and regularization do not simply destroy the bridge. At the same time, it is not a decisive model-selection victory. The right claim is therefore not “RUT wins,” but rather that the bridge remains competitive enough after tightening to justify further testing.

7.2 The H_0 issue. A second pressure point is the local expansion-rate tension. Even if the framework’s enlarged-horizon program is becoming more derived, the paper still needs a direct subsection stating whether the present workflow actually improves, worsens, or leaves unresolved the H_0 problem. The current materials are stronger on the enlarged horizon than on this point, so the honest posture is to say that the r_d program has advanced more quickly than the full H_0 reconciliation program.

7.3 The size of r_d ≈ 181.75. The enlarged horizon is a large departure from the standard expectation. That makes it interesting, but it also makes it vulnerable. The strongest current statement is therefore not “the large r_d is proven.” It is: the large r_d is no longer treated purely as an effective fit direction, because the framework now specifies a closed minimal transport-and-backreaction structure under which such an enlarged horizon can arise.

7.4 Reviewer-safe summary. A reviewer-safe framing of the present situation is: the bridge remains statistically competitive after tightening; the enlarged horizon is more mechanistically grounded than before; the framework still faces meaningful pressure from parameter economy, H_0 interpretation, and the size of the preferred horizon shift; therefore the current claim should remain narrow, empirical, and falsifiable rather than broad and triumphal.

8. Falsifiers and shared-region test

The current RUT bridge is only scientifically valuable if it can fail in clear ways. The most important next test is not more symbolic expansion. It is the **shared-region test**: determine whether one shared parameter region can simultaneously support an enlarged horizon and acceptable broader benchmark behavior.

The scan plan identifies the core parameters as transport amplitude, backreaction amplitude, preferred phase alignment, phase-coherence width, separate transport weights, mixed-sector cooperation, and the effective equation-of-state of the added backreaction sector. The point is not exhaustive realism at first pass. The point is to determine whether the closed minimal model possesses a credible shared region at all.

8.1 Enlarged-horizon falsifier. If no stable region satisfies the condition that weighted drag-era transport enhancement exceeds weighted drag-era expansion backreaction, then the operational enlarged-horizon program weakens sharply. In that case the organized sector would fail to support the very effect it was introduced to explain.

8.2 Shared-region falsifier. If regions that enlarge r_d systematically degrade the broader benchmark lane beyond acceptable tolerance, then the enlarged horizon and the broader late-time bridge would likely not be manifestations of one shared compact structure.

8.3 Transfer falsifier. If the late-time bridge’s apparent fingerprint vanishes under further cross-probe testing, or if the broad competitive basin collapses into seed-fragile behavior under tighter confrontation, then the present late-time correction reading should be weakened.

8.4 Correspondence-limit falsifier. If the model cannot preserve a clean standard limit as the added sector decouples, that would be a serious theoretical failure. One current strength of the closed minimal r_d model is that it preserves this correspondence explicitly.

8.5 Practical decision rule. Seek regions where the model yields a stable positive shift in r_d without excessive backreaction and where that same region does not degrade the broader benchmark lane beyond acceptable tolerance. In short, find whether a credible shared region exists. If it does, the next step is richer field evolution. If it does not, the present operational model must be weakened or revised.

9. Conclusion

The current strongest reading of RUT is narrower and more scientifically usable than its earlier broader formulations. In the present BAO+growth workflow, the regularized bridge is best interpreted as a statistically competitive late-time correction sector with a broad preferred basin, a repeatable growth-to-geometry transfer fingerprint, and a clear interpretation boundary. It is not yet a demonstrated replacement for standard cosmology across all redshifts, but it has survived tightening well enough to warrant compact reviewer-facing presentation.

The enlarged sound horizon is also in a stronger state than before. It has advanced from a stabilized signal near $r_d \approx 181.75$, to a structured-transport derivation path, to a first operational closed model in which the same organized sector affects both primordial transport and background expansion. That progression does not complete the early-plasma derivation, but it does materially improve the framework's posture against the criticism that the enlarged horizon is merely an effective fit direction.

The core scientific question now becomes sharper. The next threshold is no longer whether the framework can name a promising signal. It is whether one compact shared parameter region can sustain the enlarged horizon, preserve acceptable benchmark behavior, and survive broader cross-probe confrontation. The attached falsifier logic and scan plan make that next step explicit.

The fair conclusion is therefore restrained: RUT is not yet a completed unification theory and not yet a decisive cosmological replacement. But it may now be maturing into a serious late-time cosmology candidate with a partially closed enlarged-r_d program that is more derived, more constrained, and more testable than before.